

DATA/CS 172: Python for Data Analysis

Randomness

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Reminder

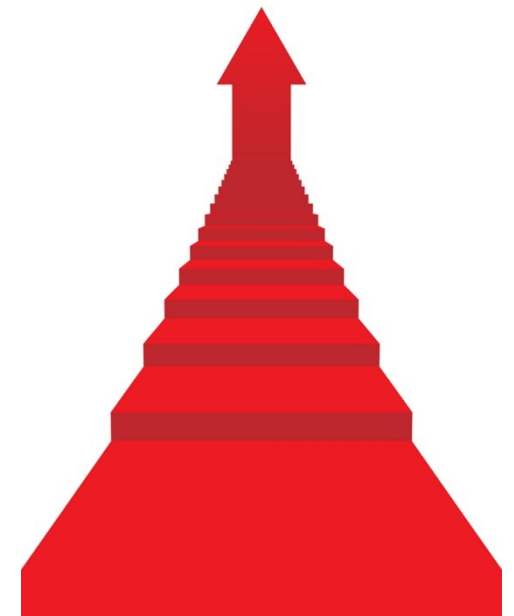
- Lab 11 due today

Agenda

- Randomness

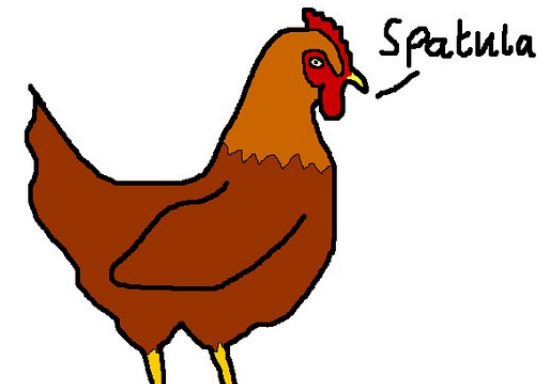
Right now our scripts are deterministic...

- Given same data, do the same thing
- Often what we want, but not always....
 - Want to test the system with synthetic data?
 - Need randomness!
 - Want to simulate a natural phenomenon?
 - Need randomness!
 - Want security and privacy?
 - Need randomness?
 - Want advanced analysis/ machine learning?
 - Need randomness!



Randomness

- The concept of randomness is incredibly important to data science
- Without the concept of randomness, statistics wouldn't exist!
- What is randomness?
 - Informally: Different stuff happens each time
 - Slightly more formal: A set of **outcomes**, each assigned a specific **probability**



Coin Example

Two Outcomes – Heads and Tails

Probability of each outcome:

Heads = 50% of the time = 0.5 probability

Tails = 50% of the time = 0.5 probability

Probabilities **must sum to 1** (100%)!



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Dice example

- Six outcomes 1,2,3,4,5,6
- Each outcome has probability 0.16666666



Getting a randomly-chosen int in Python

- `import random` ← Goes at the top of your code
- `random.randrange(a,b)`
 - Gives a number between a (inclusive) and b (exclusive)
 - Assumes all numbers have the same probability
 - Note: Much better style than `randint`
 - Demo -dice



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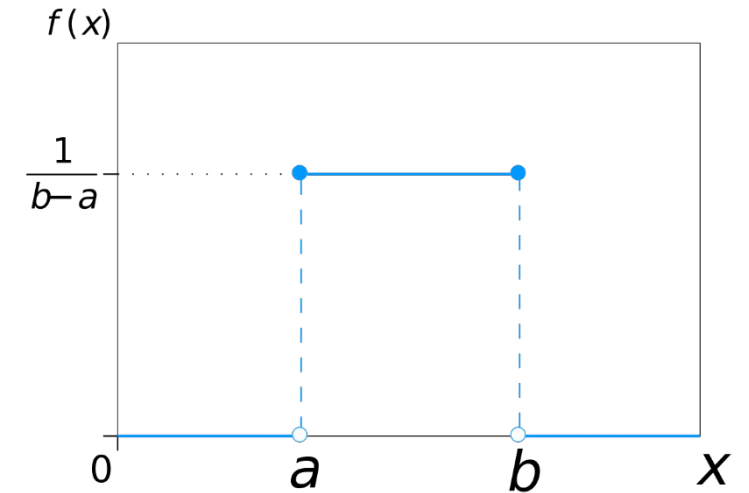
“Pick an random element from a list”

- Very common task
- Could do `rand.randrange(0,len(myList))` as use that as the index
- This is so common Python makes it easy
 - `random.choice(myList)`
- Another fun one:
 - `random.shuffle(myList)`
 - Note: does not return anything, shuffles the passed-in list



How about floats?

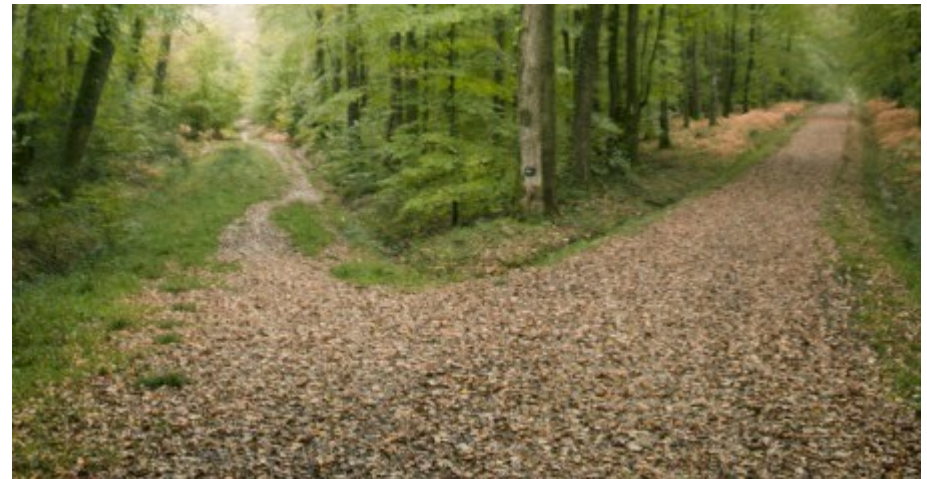
- `random.random()`
 - Probably #1 most used random function
 - Returns random float between 0 and 1
- Conceptually a bit more difficult to understand
 - Infinite set of outcomes
- Called a **uniform distribution**, - no region is weighted higher than any other
- `random.uniform(a,b)` – more general case



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“Do something with 0.8124 probability”

- Hmm... there isn't a function in random that will do this directly! ☹️
- Any ideas?
- Use `random.random()`!
- If `random.random() < 0.8124`:
 - Do something



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Final note

- Not truly random – computers are inherently deterministic!
- Called pseudorandom – to get a “random” number, takes an initial number (called a **seed**), then does a bunch of complex computations on it
 - Seed is usually clocktime (different every time)
- For testing/reproducibility, sometimes want the same every time
 - Can use `random.seed(number)`



Lab out

- Practice with randomness AND dicts of lists